

Are College Degrees Still Protected from Automation? Measuring Curricular Exposure to Artificial Intelligence

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February 23, 2026

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Abstract

Using a new dataset of course descriptions from 1,000 U.S. colleges and universities, I adapt the patent-to-task overlap method in Webb (2019) to measure how strongly each college major is exposed to three technology classes: Robots, Software, and Artificial Intelligence (AI). Exposure to Robots and Software is modest, but AI exposure is pervasive: the median major ranks around the 75th percentile of the occupation-level AI-exposure distribution. STEM and Business/Economics majors are more exposed to AI than Social Science or Humanities fields. Within a given field, teaching-focused institutions exhibit greater AI exposure than R1 research universities. Finally, I outline an event study analysis that leverages the November 2022 release of ChatGPT to estimate how student enrollment and course supply adjust to a sudden, highly salient shift in AI capabilities.

JEL Classification: J24, I2, O3.

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1 Introduction

Technological disruption has long been a feature of labor markets. Each wave of innovation expands the production possibilities frontier but may simultaneously render specific skills — and the workers who supply them — less valuable. In recent history, the brunt of automation has fallen on routine, lower-skill jobs (Autor et al. 2003). Under such skill-biased technological change, a standard policy prescription has been to encourage workers at risk of displacement to upskill through higher education or lifelong learning on the premise that higher education equips workers with harder-to-automate capabilities. That logic may now be in question.

A new generation of artificial intelligence (AI) tools differs from earlier technologies in three notable ways. First, their potential application space is extraordinarily broad, spanning language, vision, prediction, and code. Second, many models operate with minimal human supervision, performing entire task bundles rather than single modular steps. Third — and central to this paper — the tasks at which modern AI excels overlap heavily with those performed by workers in traditionally high-skill occupations (Brynjolfsson and Mitchell 2017). If AI increasingly competes on the same cognitive terrain that universities teach, the return to “upskilling” through college coursework may erode faster than in previous eras. Moreover, curricular change in higher education is notoriously sticky (Light 2024); universities may struggle to realign programs before technologies have already leapt ahead.

This paper provides the first systematic measurement of how exposed college majors are to recent technological advances, with a focus on AI. Building on the task-based approach in Webb (2019), I combine more than twenty-five years of course descriptions, enrollment counts, and offerings for 1,000 U.S. colleges and universities (Light 2024) with the universe of U.S. patents classified as Robotics, Software, or AI. For each field of study, I extract verb-object pairs (e.g., “analyze data,” “write essays”) from course descriptions, construct a task-frequency distribution, and compute its overlap with the task profiles that characterize each technology class. The resulting *exposure score* is intentionally agnostic: it captures potential for either substitution or complementarity, leaving the sign of the welfare effect for subsequent analysis.

The paper documents three headline facts. First, exposure of college courses to AI is significantly greater than their exposure to previous waves of technological change, such as Robotics or Software patents. While only one field (Skilled Trades) has exposure at or above the 75th percentile of occupation exposure to Robots, the median field has exposure to AI above the 75th percentile of the occupation-level AI exposure distribution. STEM fields — especially Statistics and Data Science — are most exposed to AI, followed by Eco-

nomics and Business. Second, AI exposure of college courses has risen steadily since 2010, even within fields of study. The growing exposure is driven largely by the incorporation of data analysis tasks into new and existing courses, both in and outside of the traditionally quantitative fields. Third, within a given field, exposure varies systematically by institution type. Teaching-oriented and regional public universities offer curricula that peak at more introductory levels and, therefore, overlap with current AI capabilities more than the comparatively advanced sequences at R1 research universities. However, the single most exposed field-institution cell is Statistics/Data Science at R1 institutions.

In preparing these results, I conduct a series of validation exercises to demonstrate that the task-based measure of human capital development in universities produces sensible and meaningful patterns. The tasks extracted from course descriptions recover intuitive relationships between fields of study and the skills emphasized in those fields. For example, tasks such as “analyze data” and “solve equations” appear most often in quantitative fields like Math and Statistics, whereas tasks such as “write essays” or “interpret texts” are prominent in English and Humanities courses. Moreover, the tasks emphasized in each field correlate with the tasks performed in the occupations graduates typically enter. When I repeat the main exposure analysis using this occupational linkage, rather than the direct field-to-technology overlap, the exposure rankings across fields are very similar. This finding suggests that the task-based approach captures a real channel through which students’ coursework aligns with the skills they deploy in the labor market.

AI will affect the broader production of knowledge through college courses in ways that are distinct from previous technology shocks. Universities have adapted to previous technology shocks: the deployment of computational tools reduces the basic arithmetic required in courses (fewer students are inverting matrices by hand), but unlocked explosive growth in coursework where students apply these techniques in analyses that would have proven prohibitively time consuming in the past; the availability of electronic library resources lowers the cost of producing research papers, facilitating an increase in project-based learning [NTD - is there evidence of this?]. Two features distinguish AI from previous coursework-augmenting technologies: first, the generative and general purpose nature of the technology indicate applications more widespread than previous innovations; second, the rapid adoption of the technology, absent the same capital constraints as in previous waves of technological change, mean that students and instructors have access to AI simultaneously and widely, such that instructors do not have the same lead time to adapt their courses based on the capabilities of the technology. Moreover, while the applications of the technology are immediately apparent to students, the implications of the technology for the kinds of human capital instructors need to offer remain uncertain.

[NTD: summary of the syllabus and grades analysis]

The paper makes three central contributions. First, it extends task-based exposure metrics from occupations to the human capital formation in college, offering a forward-looking benchmark for policy discussions around the value of college majors during a period of potential AI disruption in the labor market. Second, the analysis reveals substantial within-major heterogeneity across institution types, driven largely by differences in the level of courses offered rather than differences in skills/tasks emphasized in the same course offered across institutions. The findings demonstrate how the “same degree” can differ meaningfully depending on where it is earned and reveal limitations of data sources that record information only on an individual’s field of study. Third, the event study framework should provide new evidence on the elasticity of both student demand and course supply to fast-moving technological shocks.

The paper builds on a large literature in labor economics studying how technology affects employment and occupational choice. Autor et al. (2003) introduced the idea of modeling occupations as bundles of tasks, with technology potentially complementing or substituting for workers in those tasks. Subsequent work has shown that different technologies vary in how they alter demand for skill — some technologies complement cognitive or non-routine work, while others automate routine tasks. For example, technological change in the late 20th century was skill-biased: it disproportionately substituted for workers in jobs with lower education requirements while complementing highly educated workers, thereby contributing to the growing earnings gap between more and less educated workers (Autor et al. 2006). Although evidence on the labor-market effects of AI remains nascent because of its recent development and, as yet, limited deployment (McElheran et al. 2024), there is growing empirical support for these task-based predictions. Establishments that have adopted AI tend to demand more workers who can integrate AI into production processes, while simultaneously reducing demand for workers in occupations that AI can readily perform (Acemoglu et al. 2022).

Several recent papers have developed methods to measure the exposure of occupations to new technologies. Felten et al. (2021) and Brynjolfsson et al. (2018) rely on crowdsourced scoring: Brynjolfsson et al. (2018), for example, use a standardized rubric to assess whether tasks are suitable to machine learning based on how clearly inputs map to outputs and how easily performance can be measured. Felten et al. (2021) focus on perceptions of overlap between specific AI capabilities and occupational abilities. In contrast, Webb (2019) uses a data-driven approach: he measures exposure by comparing the text of patents (which describe what technologies do) with O*NET task descriptions (which describe what workers do). This method has several advantages for my project. First, it relies on textual similarity

between technology capabilities and tasks, rather than subjective judgments of suitability. Second, because it is based on patents rather than realized wage changes or job postings, it provides a forward-looking measure that reflects the technological frontier, rather than current market conditions. Finally, the method can be easily updated as new patents emerge, making it a flexible way to track technological change over time.¹

My paper applies this logic one step earlier in the labor market pipeline — mapping patents not to occupations, but to the tasks embedded in college courses. This shifts the unit of analysis from workers to students and moves focus from firms’ hiring decisions to universities’ curricular decisions and students’ human capital investment.

Applications of a task or skills framework to research in higher education are more limited, largely because of the need for rich data that records skills or tasks at a field- or occupation-by-field level. Recent work in this area has relied on either course descriptions or course syllabi to extract institution- and course-level detail on course content. Closest in spirit to my project is work by Javadian Sabet et al. (2024), who link course descriptions from a large corpus of syllabi at numerous colleges and universities to detailed work activities in the labor market. Their data tool offers an alternative source for assessing course or field exposure to AI, although it is limited for measuring exposure of more recent courses to technological change because its data cutoff is 2017. Biasi and Ma (2022) also use this syllabus data to track how courses incorporate material at the frontier of academic research. While this shares my goal of linking course content to cutting-edge knowledge, their focus is on courses’ inclusion of academic frontier material, whereas my analysis asks whether courses incorporate tasks that overlap with what technologies at the frontier can do. Finally, a recent white paper by Timmerman (2025) measures field-level AI exposure by mapping majors to occupations and applying occupational AI exposure scores. My approach differs in measuring exposure directly at the field-of-study level, rather than using occupations as a mediator. It also benefits from rich data that allow me to measure heterogeneity in exposure within the same field across institutions and over time.

The rest of the paper proceeds as follows. In Section 2, I summarize the data sources used for this project. Section 3 outlines the text analysis used to measure field-technology exposure. I walk through validation exercises in Section 4 to stress-test the task-based framework as applied in this new setting. Section 5 summarizes results in the measurement of field-technology exposure, and Section 6 outlines the proposed event study analysis to measure enrollment responses to the release of ChatGPT. Section 7 concludes.

¹Although I follow Webb’s use of patents here, the same methodology could also be applied to other sources of technology data, like GitHub commits or academic papers, to expand the measure’s coverage as AI evolves.

2 Data

The primary data sources for this project are patent text from the US Patent Office, covering all patents filed between 2000-2024, and comprehensive course offerings data from 1,000 US colleges and universities.

2.1 Course catalog data

The course catalog dataset contains detailed information on course offerings at a large, nationally representative panel of US colleges and universities (Light 2024). I collected these data by scraping online course catalogs and course schedules, capturing 50 million course sections offered since 1996 across 1,000 institutions.

The dataset forms an unbalanced panel that includes the complete set of course offerings for each institution-year. For each course section, I observe enrollment, instructor(s), instruction format (e.g., in-person or virtual), and a brief text description of course content. Figure 1 is an example of the information contained in a typical observation. Most crucial for this project are the course descriptions, from which I extract tasks, and the enrollment counts.

The sample is broadly representative of US colleges and universities. It covers 37% of non-profit four-year institutions and 31% of non-profit two-year institutions, including 58% and 43% of enrollment, respectively. It also mirrors national distributions of key characteristics—such as selectivity, tuition costs, and endowment levels—although it skews somewhat toward larger, public institutions because it misses many very small, often religiously affiliated private colleges.²

2.2 Syllabus data

Measuring the evolution of coursework in response to ChatGPT requires detailed information on the structure of college courses. I construct a panel of course syllabi from 20 US colleges and universities. Institutions in the syllabi sample cover a range of institutions, ranging from highly selective to non-selective four-year institutions, covering data pre- and post-the release of ChatGPT (and as early as 2005). The sample contains publicly available syllabi from universities that publish full syllabus archives online, and not behind a login wall. Essentially, I only include institutions for which as complete a set of syllabi as possible are available. The majority of the schools in the syllabus sample are located in Texas; Texas HB 2504 requires universities to make publicly available course syllabi and other course details.

²See Light (2024) for a detailed discussion of coverage and representativeness.

While the law requires archives of only two years, many institutions have archives that date back to the law’s passage in 2009.

Benchmarking against the course offerings data, the syllabus sample includes syllabi for roughly [NTD]% of undergraduate courses offered³ and [NTD]% of enrollment in these courses.

For this paper, I use an automated approach to extracting from the syllabus related to course structure and administration. In particular, I extract course identification information (course name, ID, title, and period of course offering), instructor information, grade components, and various course policies using the OpenAI API. I summarize the details of the extraction procedure, including the prompts used to extract the information, in Appendix [NTD].

For the data to be useful, it is essential that the data extraction match the information in the syllabus. Of particular concern is the risk of AI hallucinations producing output that seems plausible but is not accurate. I take a few steps to guard against this. First, I have the LLM output raw text from the syllabus in addition to the processed grade component name and weight. I can, therefore, confirm that the text output matches exactly what appears in the syllabus. Second, I validate the accuracy of the LLM output using a human evaluator based on a sample of 250 syllabi. The LLM accurately extracts the full grading scheme in [NTD]% of cases.⁴

I classify the grading components into seven categories using string matches on the raw text of the grading component: exams (tests, quizzes, etc.), homework, participation, projects, presentations, labs, and essays. I classify components that do not fit into one of the seven categories into an “other” category. I provide descriptives on the distribution of these grading components in the syllabus data in Appendix Table [NTD].

2.3 Grades data

Data on course enrollment and grades received come from [NTD].

³Excluding independent study, study abroad, private lessons, and other non-classroom based courses.

⁴Errors typically take the form of adding an extra grading component that is the total of different sub-components: for example, a syllabus may list Assignment 1 and Assignment 2 as grade components and include an Assignment Subtotal line in the grade composition table. In post-processing, I remove any grade item that includes the string “total.” These errors do not appear to be systematic for specific grading components or in specific fields or years. Thus, the consequence of these inaccuracies will primarily be to introduce measurement error into the estimates.

2.4 Patent data

I obtain patent titles and abstracts for all patents filed with the US Patent and Trademark Office between 2000-2024. Titles typically provide brief descriptions of a technology’s core function, while abstracts provide longer summaries of the technology’s development and applications. For my analysis, I use patent titles to extract the tasks a technology can perform, following Webb (2019).⁵ An example of a patent title is “Analyzing flight data using predictive models” (Desell et al. (2019)). The final patent dataset includes approximately 135,000 patents covering Robotics, Software, or Artificial Intelligence.

To classify patents into technology classes, I apply a keyword filter similar to Webb (2019). For Robots, I include patents whose titles or abstracts contain the word “robot” or derivatives of “manipulate,” excluding patents with CPC codes A61 (medical or veterinary science) and B01 (physical or chemical processes). For Software, I adapt the classification of Bessen and Hunt (2007) and include patents mentioning “software” or both “computer” and “programming,” excluding those also mentioning “circuit,” “chip,” “semiconductor,” “bus,” “antigen,” or “chromatography.”

For AI, I expand Webb’s definition to include phrases associated with recent AI innovations: “artificial intelligence,” “deep learning,” “large language model,” “llm,” “neural network,” “predictive model,” “reinforcement learning,” “supervised learning,” or “transformer model.”

The resulting dataset includes 28,165 Robot patents, 80,225 Software patents, and 26,136 AI patents. Approximately half of the patent titles contain at least one verb-object pair, resulting in around 12,000, 16,000, and 43,000 verb-object pairs for the three technology classes, respectively.

2.5 Supplemental data sources

For validation exercises, I use occupational task descriptions from O*NET, a database maintained by the US Department of Labor that summarizes worker attributes and job characteristics for nearly 1,000 occupations. These occupation details are gathered through firm and worker surveys to maintain up-to-date information. For my analysis, I use task summaries linked to occupations in the O*NET data.⁶ I use O*NET ratings of task importance and relevance to weight verb-object pairs in constructing exposure measures.

To link fields of study to occupations, I use field-occupation mappings estimated from

⁵Although abstracts offer richer detail, they often include background or development information unrelated to direct applications. Results using abstracts are qualitatively similar to those using titles.

⁶For example, an O*NET task for Economists is “Compile, analyze, and report data to explain economic phenomena and forecast market trends, applying mathematical models and statistical techniques.”

the American Community Survey (ACS). Specifically, I use ACS surveys from 2009-2019 and restrict to employed workers ages 25-65 who hold a four-year degree. This mapping serves as an external validity check for my task-based exposure measures, allowing me to compare the skills emphasized in college courses to those performed in jobs linked to these courses.

3 Methodology

The core methodology for this project involves three main steps: first, extracting “tasks” (verb-object pairs) from course descriptions and patent titles; second, constructing distributions of these tasks for each field of study and technology class; and third, calculating an overlap statistic that reflects the exposure of each field (or field-institution cell) to each technology. This exposure score captures the degree to which the core tasks students perform in their courses are also performed by new technologies.

3.1 Defining and extracting tasks

Text data encode rich information, but the core challenge is reducing this high-dimensional, interconnected content into a tractable form that captures what is economically meaningful. In this project, I must extract from course descriptions — documents that include administrative, pedagogical, and contextual details — only the parts relevant for measuring a course’s technology overlap. A similar process occurs, implicitly, in most economic analyses: for example, researchers often control for an individual’s race, gender, or age in labor studies, but not for their quarter of birth or year of marriage unless relevant. With text data, the separation between relevant and irrelevant information is harder to anticipate, so applications typically rely on data-driven approaches to isolate the pieces of text most pertinent for analysis.

I define a *task* as a verb-object pair. For example, consider the description for Stanford’s COMM 274D (Public Affairs Data Journalism II): “Learn how to find, create and analyze data to tell news stories with public service impact. Uses relational databases, advanced queries, basic statistics, and mapping to analyze data for storytelling...” (Stanford 2025). The description includes multiple verb-object pairs: “find data,” “create data,” “analyze data,” “tell stories,” “use databases,” “use queries,” “use statistics,” and “use mapping.” These pairs provide a concise representation of the main activities taught in the course. The motivation for the analysis in this paper is that if patents for new technologies also perform these tasks, the course content itself (and, therefore, the human capital students develop in the course) may be exposed to potential automation or augmentation by that technology.

Given the large size of the text corpus and the complexity of natural language, manual

review is not feasible. Instead, I use common Natural Language Processing (NLP) techniques to clean and parse the text at scale. I pre-process the text by removing punctuation and standardizing capitalization. Next, I extract verb-object pairs using the `spaCy` package in Python, a widely used NLP tool (Honnibal et al. 2020). I then remove pairs in which the verb or object is a stopword⁷, and lemmatize the verb and object to a common form (e.g., “analyzing” and “analyze” become “analyze”).

Course descriptions are noisier than patent titles or O*NET tasks, as they often include administrative details or generic phrases that are not relevant for measuring the human capital students develop in these courses. To reduce noise, I first remove “boilerplate” sentences (such as prerequisites and enrollment restrictions). Next, I calculate chi-squared statistics for each verb-object pair, comparing their occurrence frequency in course descriptions versus patents. I drop pairs that are among the 100 pairs with the highest chi-squared statistic, suggesting that these pairs are broadly related to course descriptions as a document type rather than the specific tasks students perform or human capital they develop.⁸

To further ensure that the verb-object pairs are closely related to course tasks, I project the cleaned task dictionary onto the union of task pairs that appear in either the patent data or the O*NET task inventory (excluding previously dropped pairs). This step ensures that exposure scores only reflect tasks that are relevant to technological change or to job tasks more broadly.

The final dictionary includes 27,612 unique verb-object pairs. Table 3 summarizes how many pairs are dropped at each step. In Section 4, I summarize validation exercises, including manual review of top pairs by field, to confirm that the resulting task dictionary captures the main activities taught in those courses.

3.2 Exposure measure

I calculate the exposure of a field of study s to a technology class t as the frequency-weighted average of the tasks students perform in that field, weighted by the importance of those tasks to the technology. Let D denote the dictionary of verb-object pairs d . For field s , the share of task d is:

$$x_s(d) = \frac{\sum_{\delta \in D_s} \mathbb{I}(\delta = d)}{|D_s|} \quad (1)$$

where D_s is the set of all task observations in field s . For example, in Economics courses offered in 2023-24, “analyze data” accounts for 1.8% of all verb-object pairs.

⁷For example, “use” or “have,” which lack meaningful economic content (Jurafsky and Martin 2014).

⁸For example, “introduce student” or “receive credit.” Examples of dropped pairs are listed in Appendix Table 2.

Similarly, for each technology class t , I compute the share $w_t(d)$ of task d among all tasks mentioned in patents for that technology:

$$w_t(d) = \frac{\sum_{\delta \in D_t} \mathbb{I}(\delta = d)}{|D_t|} \quad (2)$$

The exposure of field s to technology t is then the weighted average of task shares:

$$exposure_{s,t} = \sum_{d \in D} x_s(d) w_t(d) \quad (3)$$

Intuitively, this measure captures how much of the “task footprint” of a given field overlaps with the tasks that a given technology is already able to perform. Higher values of $exposure_{s,t}$ indicate that a larger share of the tasks emphasized in a field’s coursework are tasks that new technologies claim to be able to do. In later analyses, I extend this exposure calculation to field-institution cells to account for differences in course sequencing and teaching emphasis across different types of colleges and universities.

4 Validation

I conduct a series of validation exercises to confirm that the task-based exposure measures capture meaningful differences across fields of study and align with intuitive and observable patterns in the data.

4.1 Distinctive tasks by field of study

To demonstrate how the task extraction approach captures variation across fields, Table 1 summarizes the most common verb-object pairs for a representative sample of fields. For each field, I select the three verbs that appear most frequently and, for each verb, list the (up to) four most frequent objects.

The table highlights distinctive and intuitive task patterns across fields. For example, English courses emphasize tasks such as “reading texts” and “writing essays,” Education courses emphasize “teaching lessons” and “developing curriculum,” and Arts courses emphasize “creating images” and “performing music.” At the same time, some tasks — particularly those involving data analysis — appear across multiple fields, reflecting the growing role of data skills across a range of disciplines.

Columns 5-7 of Table 1 show the technology exposure of these most important tasks, calculated as a weighted average of task frequency in course descriptions and task relevance to each technology class. For readability, I multiply the scores by 100. The table reveals

that data analysis tasks are highly exposed to AI across the board, with particularly high exposure for fields such as Economics and Statistics/Data Science where this form of analysis is emphasized. More broadly, exposure to Robots is generally the lowest across all fields, followed by Software. Where tasks overlap with AI, the extent of that overlap is often much greater than for other technology classes.

4.2 Linking field tasks to occupation tasks

As a second validation exercise, I assess whether the tasks extracted from course descriptions for a given field also appear in the occupations into which graduates from that field typically flow. This provides an external check that the task-based measure of field exposure captures tasks that matter in the labor market.

To do so, I use field-to-occupation mappings developed from the ACS for prime-age workers (ages 22-65).⁹ I then construct task vectors for each occupation and each field of study, based respectively on the importance of verb-object pairs in O*NET and their share in course descriptions. For this exercise, I restrict the task space to only those pairs that appear in any O*NET task descriptions. I measure the similarity between each field-occupation pair by calculating the cosine distance in task space. To ensure that I focus on meaningful matches, I drop any field-occupation pair with either an empirical share below 0.01% in the ACS and no task overlap.¹⁰ Finally, I compute the rank-rank correlation between the empirical field-to-occupation shares and the task-based rankings.

The resulting rank-rank correlation is 0.31, indicating a meaningful positive relationship between the two measures. Given the sparsity of O*NET task lists and the fact that some fields map to multiple occupations imperfectly, this relatively large and positive correlation validates the relevance of my task-based measure. It also provides new evidence of how the tasks students learn in college align with the jobs they enter after graduation.

5 Results

5.1 Overall overlap between courses and technology

Figure 2 plots the distributions of technology exposure scores for each of the three technology classes (Robots, Software, and AI) using course descriptions from the 2023-2024 academic year. For each of 52 fields of study (e.g., Math, English, History), I calculate the average

⁹A robustness exercise that uses only early-career workers (ages 25-35) produces similar field-to-occupation mappings.

¹⁰Dropping these occupation-field pairs is generally conservative for this validation exercise, as it excludes pairs where the occupation-to-field shares and task overlap are strongly correlated.

exposure score by pooling tasks from all undergraduate-level courses offered across institutions.

The results indicate that most fields of study have low exposure to Robots, with some modest variation. The field most exposed to Robots is Skilled Trades, which includes courses in vocational subjects like Construction and Auto Manufacturing, mainly offered at community colleges. Computer Science, Medicine, Engineering, and Agriculture also have relatively higher exposure to Robots, though the absolute levels are low. Similarly, exposure to Software is generally low across fields, with Library Science showing the highest overlap, followed by Statistics/Data Science, Computer Science, Engineering, Business, and Economics. However, overall exposure to both Robots and Software is minimal across most fields.

In contrast, the AI exposure distribution is wider and shows substantially higher values for almost every field. The most exposed field is Statistics/Data Science, with exposure scores nearly five times the average across fields. Other highly exposed fields include Linguistics, Computer Science, Economics, and Chemistry. Generally, STEM and Business fields are more exposed to AI, followed by Social Sciences, while Humanities fields (such as History, Ethnic/Cultural Studies, American Studies, and Religion) show the lowest exposure.

For reference, in each panel I plot the 75th and 95th percentiles of the occupational technology exposure distribution, using the Webb (2019) scoring.¹¹ For Robots and Software, most fields of study score well below the 75th percentile of occupational exposure. In contrast, the median field scores just above the 75th percentile for AI, and the most exposed field (Statistics/Data Science) exceeds the 95th percentile of occupational exposure to AI. This pattern is consistent with prior work showing that Robots and Software overlap more with low-skill jobs, whereas AI overlaps more with higher-skill activities.

There are two primary takeaways from this figure. First, the same fields that were most exposed to previous waves of technology through this task-based measure are the fields most exposed to AI. This may be due to overlapping task intensity across the three technologies (for example, ‘analyze data’ is a task common across the three technologies, potentially due to misclassification, but is substantially more important for AI than other technology categories). However, second, AI exposure is markedly higher across almost all fields compared to previous waves of technological change, especially when compared to

¹¹Projecting the course verb-object pair dictionary onto the union of the patent and occupation pair dictionaries produces a distribution of exposure scores for courses that is similar in magnitude to the occupation exposure scores. However, some of the pairs that appear in course descriptions (particularly those related to administrative details) do not overlap with on-the-job activities. Projecting only onto the patent and occupation verb-object pairs thus introduces an upward bias in the exposure scores. As a result, what matters for interpretation is not the absolute level of these scores, but rather the relative scores across fields within a given technology class, and how field-level exposure compares to different percentiles of the occupation exposure distribution.

occupation-level exposure to these technologies.

5.2 Time trend in technology exposure

Figure 3 shows how exposure to each technology has evolved over the past two decades. I estimate these trends by regressing field-technology exposure scores on year dummies, controlling for institution-by-field fixed effects. This approach isolates changes within a field of study rather than differences driven by compositional changes across fields.

The figure reveals a steady increase in AI exposure across nearly all fields since approximately 2012-13. On average, courses across fields have increased in their AI exposure by 0.00003 points (equivalent to 0.5 standard deviations in the occupation-AI exposure distribution) over the last decade. This trend is most pronounced in fields that have incorporated more data analysis tasks into their curricula. In contrast, exposure to Robots and (to a lesser extent) Software has remained largely flat over time, suggesting that the growing AI exposure reflects the specific expansion of AI-overlapping tasks into a broader set of courses.

Although there is no clear exogenous shock in 2012-13 to explain the inflection point, this timing coincides with broader shifts in student demand toward Computer Science and STEM majors and away from Education and Humanities (Light 2024). Universities may have responded by incorporating more data-intensive content into their courses to align with these demand shifts, indirectly driving up AI exposure across multiple disciplines.

5.3 Heterogeneity by institution type

Institutions differ in the types of courses they offer and how they structure those courses, which may influence patterns of technology exposure. From a policy perspective, understanding this heterogeneity can identify which types of institutions and students are most exposed to growing AI capabilities. Additionally, the results in this section provide new insight into how instruction in the same field of study differs at different types of institutions.

I calculate exposure scores separately for field-technology-institution type cells. I classify institutions into four categories based on Carnegie classifications and degree offerings: R1 universities (very high research activity), R2 universities (high research activity), teaching-focused four-year universities, and two-year community colleges. For each cell, I pool course descriptions to calculate task distributions and measure their overlap with the patent-derived task distributions for each technology.

Figure 4 plots the distributions of exposure scores across these institution categories. Exposure to Robots and Software is uniformly low across all types of institutions, with R1 universities generally having the lowest exposure to these technologies. In contrast, AI expo-

sure has greater variation across institution types and fields. On average, teaching-focused universities have higher AI exposure than both research universities and community colleges. This higher exposure is not due to differences in how the same courses are taught at these institutions — for example, the AI exposure of Principles of Economics is similar regardless of institution type. Instead, the difference arises from the types of courses offered. Teaching-focused universities often offer introductory-level courses and lack the advanced electives more common at research universities. For example, among teaching-focused universities in my sample, 75% offer pre-calculus courses and only 65% offer advanced courses such as Real Analysis. At R1 universities, these shares are 60% and 94%, respectively. This suggests that as AI’s capabilities move up the ladder of cognitive tasks, they will match the level of courses at teaching-focused universities before reaching the more advanced material taught at research universities.

Finally, the single most exposed field-institution cell is Statistics/Data Science at R1 universities, followed by the same field at R2 universities. The creation of new Data Science programs and the broader expansion of applied statistics courses appear to drive this high level of AI exposure.

6 ChatGPT event study

Having described exposure to automation broadly, this section provides evidence of how universities and students have responded to recent advances in the availability and capabilities of technology.

On November 30, 2022, OpenAI launched a prototype of its Large Language Model-based chatbot, ChatGPT. Adoption of ChatGPT accelerated rapidly: ChatGPT acquired 100 million users in shortly more than a month after release and a billion users within [NTD].

The applications of ChatGPT for higher education were [NTD].

6.1 Conceptual framework

How should instructors and students respond to the availability of new technology? I outline a brief conceptual framework below.

Students select courses to maximize expected return on investment in higher education, subject to university-imposed constraints (e.g., distribution requirements). Returns can come in the form of future earnings through the human capital accumulated in their courses or non-financial returns (artistic expression, civic exploration, or pure consumption) which may materialize over a short or long term. Taking courses is costly, both in financial costs of tuition and in psychic costs related to the difficulty of taking courses Arcidiacono (2004).

While the tuition costs of courses do not vary widely across programs, From a sample of 165 public universities, Stange (2015) demonstrates that approximately 40% charged higher tuition for at least one program. Many of these programs are in Engineering, Business, or Nursing. In programs that charge supplemental fees, the average surcharge is 10%, with substantial variance (Nelson 2008). In the course offerings data analyzed in this paper, surcharges for a course are often in Business or lab science courses and related fees are typically around \$100. but the cognitive costs of course taking may vary substantially across students and fields. Indeed, Arcidiacono (2004) finds that these costs are a primary driver of ability sorting across fields of study.

A technology innovation like ChatGPT has two, potentially offsetting, impacts on a student's objective function that may influence course selection. Immediately, labor-augmenting technology reduces the cost of taking courses. Technology that makes students more efficient reduces the amount of time and cognitive effort students need to expend on their coursework. In the extreme, and as indicated in the previous section, AI tools may be capable of substituting for parts of student labor in ways that are distinct from previous technological changes. The implication is that cognitive cost becomes less of a barrier to entry into challenging courses; students can earn higher grades and pass technology-exposed courses that would have been prohibitively difficult prior to the technology shock. The result is a push into courses linked to higher earnings.

A technological innovation also affects expected returns in the labor market. The same skills that can be augmented, or automated, in their development in the classroom by a new technology may become less valuable (or the expected value may be changed) as a result of the new technology. In other words, the same technology that can assist in a student's course can assist in their job; the same technology that can do parts of their coursework may also be able to autonomously perform parts of their job. Depending on the degree of substitutability, the effect may be a reduction in the expected value of technology-exposed human capital [NTD - cite literature]. Even if the degree of substitutability is not known, the technology shock generates greater uncertainty regarding the expected returns to skill development.¹² A reduction in the expected return to highly exposed skills will push students away from highly exposed classes.

In partial equilibrium, where course offerings hold constant but students respond to the technology shock, student grades should increase.¹³ The new AI tool increases productivity,

¹²Commentary from students and labor market observers suggests that many expect technology to lead to a reduction in the expected return to these skills, through a reduction in the number of high-earning openings in highly exposed fields. Expectations among economists are more cautious; however, but there does not appear to be widespread evidence that students or observers expect AI to increase the expected return on highly exposed skills.

¹³Barring an adjustment period where students learn the capabilities of the technology and adjust uses to

particularly in courses where skills developed align with AI’s capabilities and outputs resemble outputs that AI can produce. The net effect on course selection is ambiguous. Courses that are highly exposed to AI will become less costly for students who would prefer to take the courses but who find the coursework prohibitively difficult. However, by introducing uncertainty in the expected return to skills developed in these courses, enrollment may shift from highly exposed courses to less exposed courses, where earnings uncertainty is less impacted. Perversely, the net effect is higher adverse selection into highly exposed courses from students with lower aptitude or stamina in these courses (high cost) and are drawn by higher earnings, and reduced enrollment for higher aptitude but risk averse students.

Instructors seek to equip students with human capital at minimal time cost, subject to constraints on the course/field’s production technology and institution- or field-imposed constraints on course flexibility. [NTD - outline the instructor problem]

New technology affects the instructor’s objective function by reducing the amount of human capital, independent of the technology, that a student will develop (or, at least, reducing the signal of human capital development). When students have the capability to produce work that demonstrates not their independent abilities but work that may have entirely been done by the new technology, instructors will negatively update expectations on the amount of human capital students are developing (Akerlof 1978). Instructors can introduce policies prohibiting the use of AI tools in their courses, but the nature of the technology — to produce work that is indistinguishable from human output — complicates the enforcement of such a policy.

Instructors have two strategies to combat the effects of the new technology. First, instructors can modify the structure of their courses incentivize students’ demonstration of human capital without the assistance of the new technology. Certain grade components — homework, essays, and out-of-class projects — can be more easily performed with the assistance of AI than other components — exams, in-class participation.¹⁴ Instructors can get a stronger signal of students’ knowledge, and incentivize independent human capital development, by increasing the relative contribution of less-exposed evaluations and reducing the contribution of more-exposed evaluations. Second, instructors can change the human capital students develop in their courses, either by modifying existing courses or creating new courses, that more explicitly structure student learning in a way that is at most complemented by, rather than substitutable with, the new technology. Institutional policies may limit instructors’ ability to pursue this second strategy: multi-year planning cycles may limit institutions’ ability to

settings where the technology is labor-augmenting.

¹⁴Anecdotal evidence suggests that this is an important form of adaptation. Orders of Blue Books, used to take written exams in person, from some large universities have increased substantially since 2022 (Cohen 2025).

hire new instructors, course review processes may delay the deployment of new courses or modifications to existing courses.

Taken together, the framework yields a few predictions in partial equilibrium for how instructors and students may adapt to the deployment of a coursework- and labor-augmenting technology. First, students will use the new technology to augment (or, in some cases, substitute for) their labor. As a consequence, grades will increase. Effects on enrollment are ambiguous, but adverse selection into highly exposed courses will increase. Second, instructors will be slower to respond, and will respond more on the intensive margin (modifications to existing classes) than extensive margin due to administrative challenges and broader uncertainty about the forms of human capital that students need to develop in a technology-shocked environment.

6.2 Adaptation of course offerings to ChatGPT

In subsequent sections, I test empirically predictions of the conceptual framework outlined in the previous section. I first analyze instructor responses to the release of ChatGPT, then analyze changes in course enrollment and grades following the release of ChatGPT. I bring the two analyses together to characterize drivers of changes in grades since the release of ChatGPT. All analyses take place in an event study framework around the release of ChatGPT in November 2022 (late in the Fall 2022-23 semester).

I assess responses of course supply to the release of ChatGPT using a panel of syllabi from a sample of 20 institutions, covering the pre- and post-ChatGPT periods. From the syllabi, I extract the weights instructors assign to each grade component.¹⁵ The conceptual framework predicts that instructors responding to the release of ChatGPT will adjust course evaluations to favor forms of evaluation that students cannot complete with the assistance of ChatGPT (e.g., in-person exams, presentations, participation) and de-emphasize evaluations that can be done with/by ChatGPT (e.g., essays, homework, projects). The primary distinction between these classes of evaluations is simply whether they are conducted during our outside of class.

To test whether instructors are adapting their courses, I estimate event study regressions of grading weight before and after the release of ChatGPT. The regression takes the form:

$$y_{i,s,t}^q = \alpha_{i,s}^q + \sum_{t \in T} \beta_t^q + \varepsilon_{i,s,t}^q$$

where $y_{i,s,t}^q$ is the weight on grade component q (exam, homework, essay, etc.) in course i at

¹⁵Refer to Section 2.2 for a description of the data and grade component extraction procedure.

institution s in semester t . The parameters of interest are the event study β_t terms, which capture the change in weight on component type q relative to omitted period Fall 2022. Controlling for course fixed effects isolates variation within-course on the forms of evaluation used. Observations are at the course-by-semester level, and standard errors are clustered by course.

Figure 5 plots the event study coefficients. Starting with the Fall 2020 academic year (the first term that began after start of the Covid-19 pandemic), course weights are pretty stable. However, following the release of Chat-GPT at the end of the Fall 2022 semester, the figure documents a gradual decrease in the the essay component of courses, a small increase in the share of grades covered by homework, and modest increases in participation and presentation. The decrease in essays and increase in presentation and participation components align with predictions that instructors will adjust course grades to emphasize in-class evaluation and reduce the weight on tasks that can be performed by the AI tools. The increase in weight on homework runs contrary to predictions but may reflect a longer-term trend in the increasing weight on homework pre-dating the pandemic. The adjustments are gradual in the semesters following ChatGPT’s release. As usage increases and the capabilities of the LLM tools increases, instructors make more modifications to their courses.

To contextualize the estimates: by Fall 2025, the change in weight on essays represents a 11% decrease from the Fall 2022 baseline. The increasing emphasis on presentations and participation represents a increase from the Fall 2022 baseline.

Fields differ in their emphasis on AI-exposed forms of evaluation at baseline. Essays represented 20% of grades in the average course in a Humanities field, compared to just 4% in Computer Science courses. Presentations ranged from 5.5% of grades in Education courses to 1.6% in Computer Science courses. These baseline differences create useful variation: instructors who choose to modify their courses in response to the release of ChatGPT vary in how much they’d need to make adjustments due to the inherent production technology of human capital in their courses at baseline.

Figure 6 splits courses by field group to assess whether instructors differentially modified their courses in response to ChatGPT’s release. The figure reveals [NTD].

Figure 7 splits institutions by selectivity. The figure demonstrates that shifts in the grade composition at selective universities are larger than at less selective universities. The weight on ChatGPT-exposed tasks decreases by 1.5 percentage points ([NTD]%), but is essentially unchanged at less selective universities. The finding aligns with previous work suggesting that less selective universities offer courses that change less dynamically and are less aligned to changes in the labor market (Light 2024).

Changes in instruction technology may understate instructors’ recognition and concern

about the availability of AI tools as a substitute for students’ human capital development. Figure 8 plots an event study around ChatGPT’s release documenting changes in the share of syllabi containing an AI policy around the release of ChatGPT. Prior to Fall 2022, very few syllabi included any AI policies.¹⁶ Many of the Spring 2023 syllabi were written before the widespread adoption of ChatGPT, so the immediate increase in the share of syllabi with an AI policy is initially quite modest. By Fall 2025, more than half of course syllabi include an AI policy. Most of these policies restrict students’ use of AI tools, either in accordance with a university-wide policy or to align with instructor preferences.¹⁷

[NTD - quick discussion]: The results in this section suggest that instructors responded to the release of ChatGPT by decreasing the emphasis on essays and increased the emphasis on participation, presentation, and homework. The changes occurred gradually over the years following ChatGPT’s release. The findings demonstrate how instructors adapt to changing technology. The results also have implications for the skill composition of the skilled labor force; reducing the weight on essays results in cohorts of students with diminished writing skills.

6.3 Course taking and grades

- The conceptual framework suggests that, holding course organization fixed, a technology shock reconfigures student enrollment and grading across courses.
- In this section, I use transcript data to characterize changes in average grades received in college courses surrounding the release of ChatGPT. Figure 9 plots the trend in overall grades received over the last 15 years across Texas’ public universities. The figure highlights three issues that will complicate identification. First, the period 2010-2018 saw a secular increase in grades, part of a well-documented trend in grade inflation across US universities ([NTD: cites]). Second, average grades increased dramatically during the Covid-19 pandemic. Relative to the pre-pandemic baseline, grades increased by a third of a point (equivalent to incrementing up every student’s grade in every course by one increment, such as from a B to a B+).¹⁸ Third, owing to pandemic

¹⁶Syllabi flagged as having an AI policy prior to Fall 2022 are mostly foreign language classes that restrict the use of online translation tools. Prior to the release of ChatGPT, fewer than 1% of courses had an AI policy in the syllabus.

¹⁷Appendix Figures 10 and 11 plot heterogeneous adoption of AI policies, splitting by field of study and institution selectivity. Widest adoption of AI policies occurs in Social Science courses, while adoption initially lagged in STEM fields. Similarly, initial introduction of AI policy language was greater at selective compared to less selective institutions. However, the gap erodes by Fall 2023.

¹⁸Texas schools resumed in-person instruction during the Fall 2020 semester, so grade inflation during the 2020-21 academic year does not reflect changes related to instruction technology, but likely other accommodations/leniency from instructors during the year most affected by the pandemic.

learning loss or changes in admissions policies during the pandemic (such as test-optional admissions), the capabilities of student cohorts pre- vs post-pandemic may differ.

- I address these challenges by [NTD].
- Question 1: do changes in enrollment across courses or average grades differ by ChatGPT exposure?
- Question 2: do changes in enrollment across courses or average grades differ by the share of the grade in ChatGPT-exposed evaluations?

7 Conclusion

This paper adapts the approach of Webb (2019) to provide new measures of the exposure of college courses to emerging technologies, including Artificial Intelligence. Whereas Webb maps patents to O*NET tasks to identify which jobs are technologically exposed, I map the same patents to the tasks embedded in college courses and majors. This shift moves the focus from labor demand adjustment to the upstream investment (and potential misallocation) of human capital.

The descriptive results confirm predictions in the literature that AI is distinct from earlier technological waves in the degree to which it overlaps with high-skill tasks. Exposure of college courses to Robots and Software is generally low, but the median field of study has AI exposure comparable to the 75th percentile of the occupation-level AI exposure distribution. STEM and Business/Economics courses are the most exposed to AI, as are courses at teaching-focused universities compared to research institutions.

Next steps will focus on the event study analysis outlined in Section 6. A central challenge is the identification assumption of parallel enrollment trends for highly versus less exposed courses within the same field — an assumption that may be complicated by Covid-19 related enrollment shocks in the immediate pre-period. If this assumption does not hold in the data, alternative strategies will be needed. Additionally, I plan to develop a simple conceptual model to formalize the competing forces of technology-induced demand and substitution in course-taking decisions. Finally, to move closer to a welfare calculation, I will explore back-of-the-envelope estimates of how the concentration of AI exposure in high-earning fields could compress college major-earnings gaps.

More broadly, an open question is whether (and in what ways) task-technology overlap indicates a risk of substitutability versus an opportunity for productivity gains. I remain intentionally agnostic in this draft about whether exposure implies a threat of automation

or a complementarity that could enhance human capabilities. This question is first-order for interpreting the policy implications of these findings and represents an important direction for continued research.

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Figures and Tables

Figure 1. Example course

ECON 101

Lecture: 01

Units: 5

Class#: 7559

Economic Policy Seminar

Department of Economics [↗](#)

Lecture, Discussion

1/5/26 - 3/13/26

 Mon, Wed

 1:30 PM - 3:20 PM

 [Econ 140](#)

Instructor: Light, J.

Enrollment Status

Open Seats: 7

Enrolled: 17 Waitlist: 0

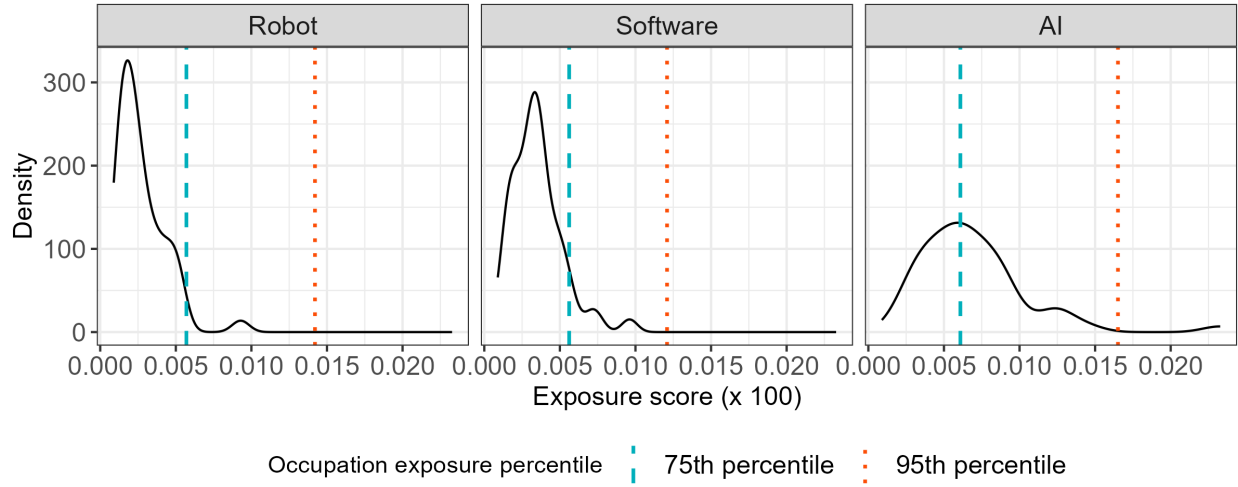
Capacity: 24 Waitlist Max: No waitlist

Course Description

Capstone and writing in the major course open to Econ majors only. Economic policy analysis, writing and oral presentations will be large components of this course. Students may also complete group projects that include empirical economic analysis focused on a specific topic. The goal of this course is to enable students to utilize the skills they have acquired throughout their time in the major. Section topics vary by instructor. Enrollment limited to application at the start of each school year with student placement notifications before the term starts. Permission numbers will be provided to students. Limited to students applying to graduate in 2025-26. Enrollment by application: <https://economics.stanford.edu/forms>.

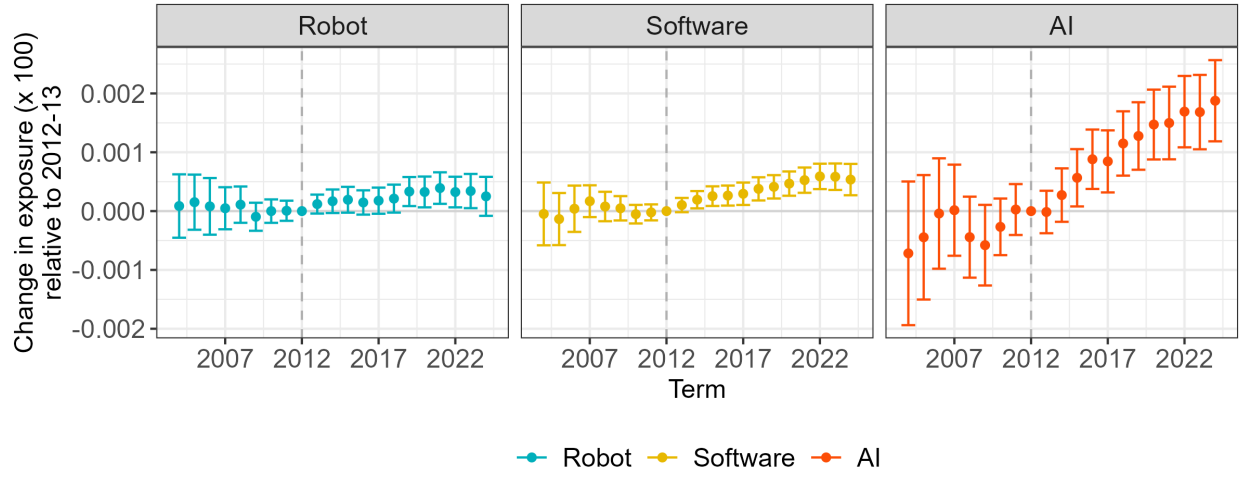
Source: Stanford University.

Figure 2. Technology exposure distribution across fields



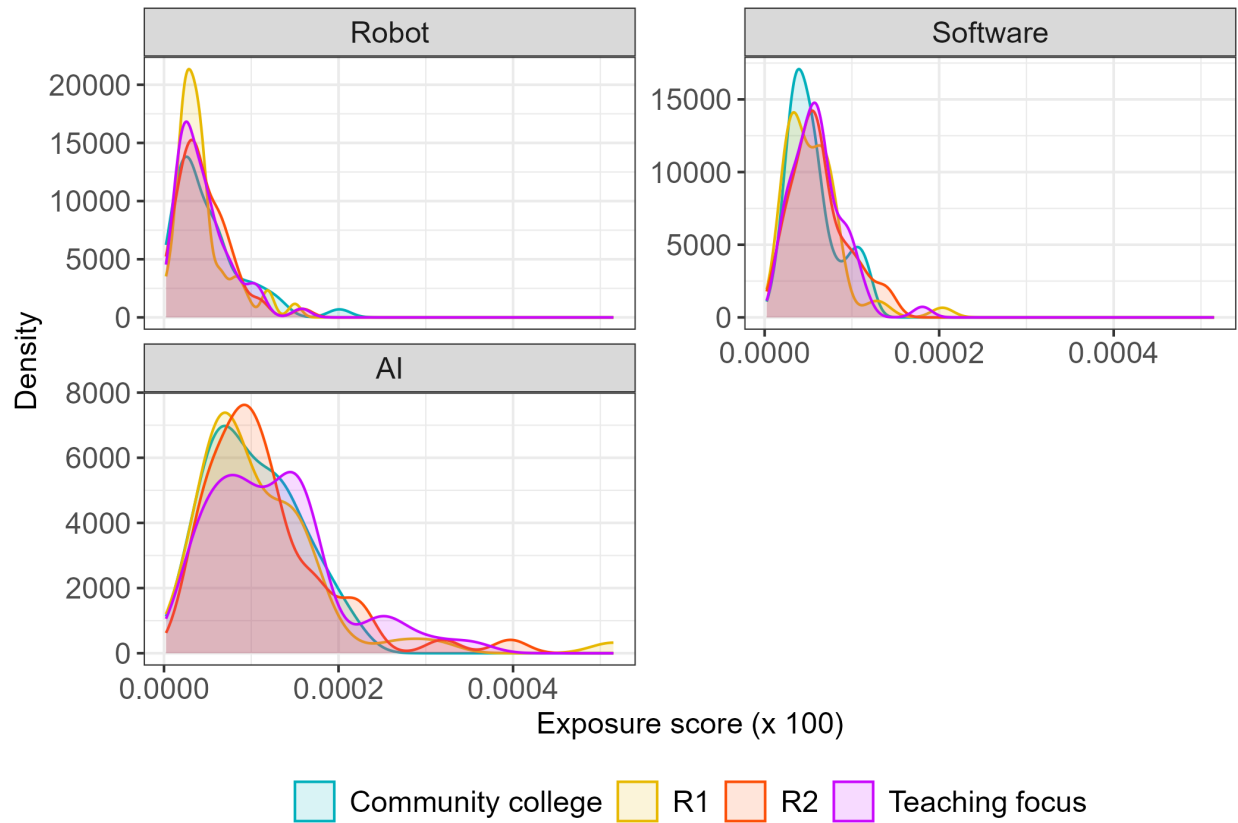
Notes: The figure plots density plots of field-technology exposure scores, split by technology class. Courses are partitioned into 52 fields (e.g., Math, Business, History). Vertical lines represent the 75th and 95th percentiles of the corresponding occupation-technology exposure distributions.

Figure 3. Technology exposure trend across fields



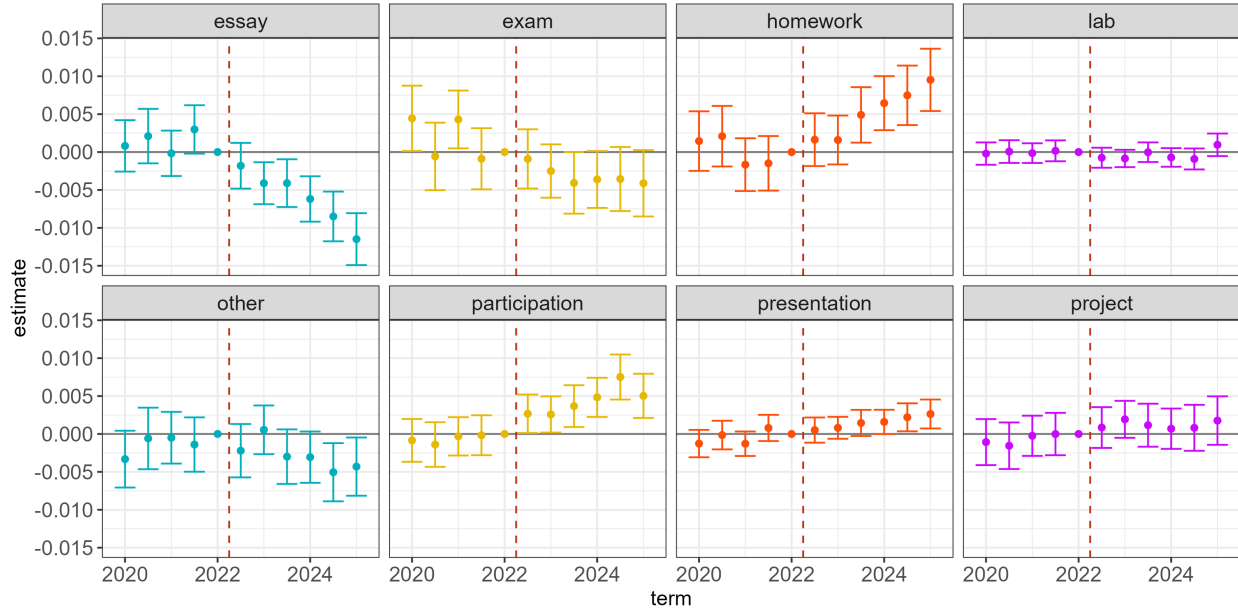
Notes: The figure plots the time trend in average technology exposure by field. Estimates come from separate regressions of technology exposure score on a vector of time dummies, controlling for institution-by-field fixed effects. Observations are at the institution-field-year level, and standard errors are clustered at the institution-field level. Estimates are relative to the average field-technology exposure in 2012-13.

Figure 4. Technology exposure distribution across fields, by institution category



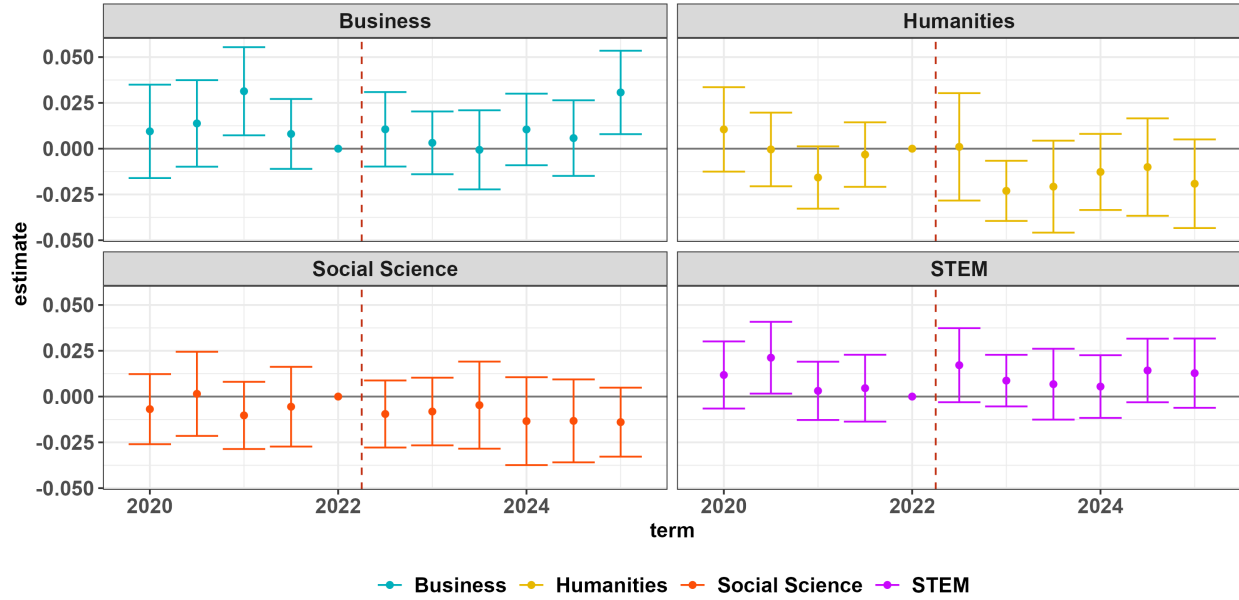
Notes: The figure plots overlapping density plots of field-technology exposure scores, with institutions broken out by degree level and research intensity (Carnegie classification). Exposure scores are calculated using course descriptions for the 2023-24 academic year. Within each school category-field cell, course descriptions are pooled together and verb-object pairs are extracted to calculate separate verb-object pair distributions.

Figure 5. Changes to assessment weights around release of ChatGPT



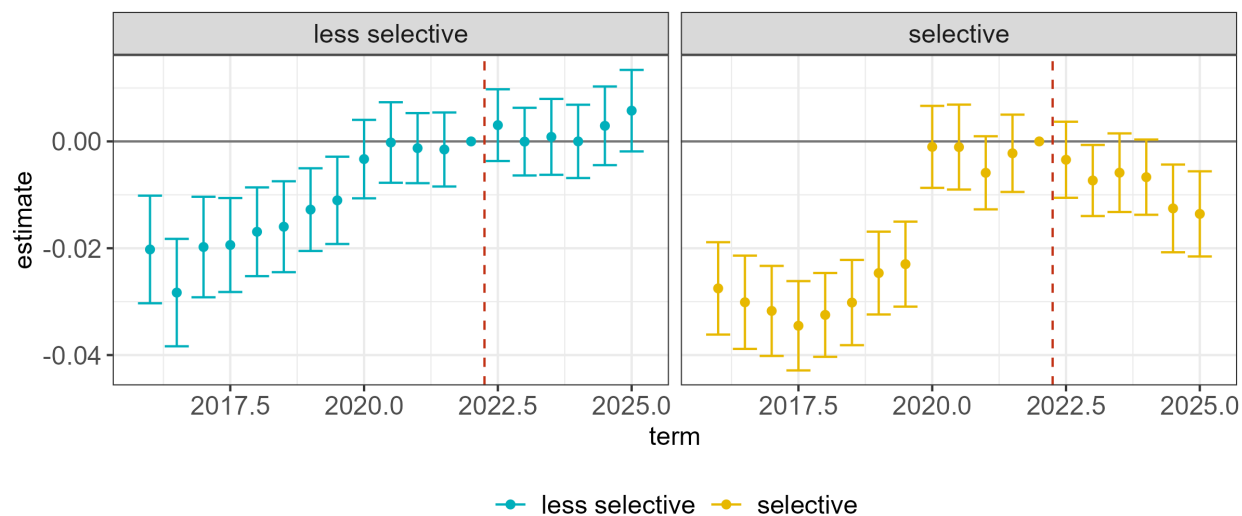
Notes: Grading weights from a sample of syllabi from 20 US colleges and universities. Estimates from an interrupted time series regression, controlling for course fixed effects. Fall 2022 is the omitted term. Time variable in academic years (thus, the 2022 point refers to the Fall 2022 semester, the 2022.5 point refers to Spring 2023, etc.). The dashed line indicates the public release of ChatGPT. Standard errors clustered at the course level.

Figure 6. Heterogeneity in grading weight changes by field of study



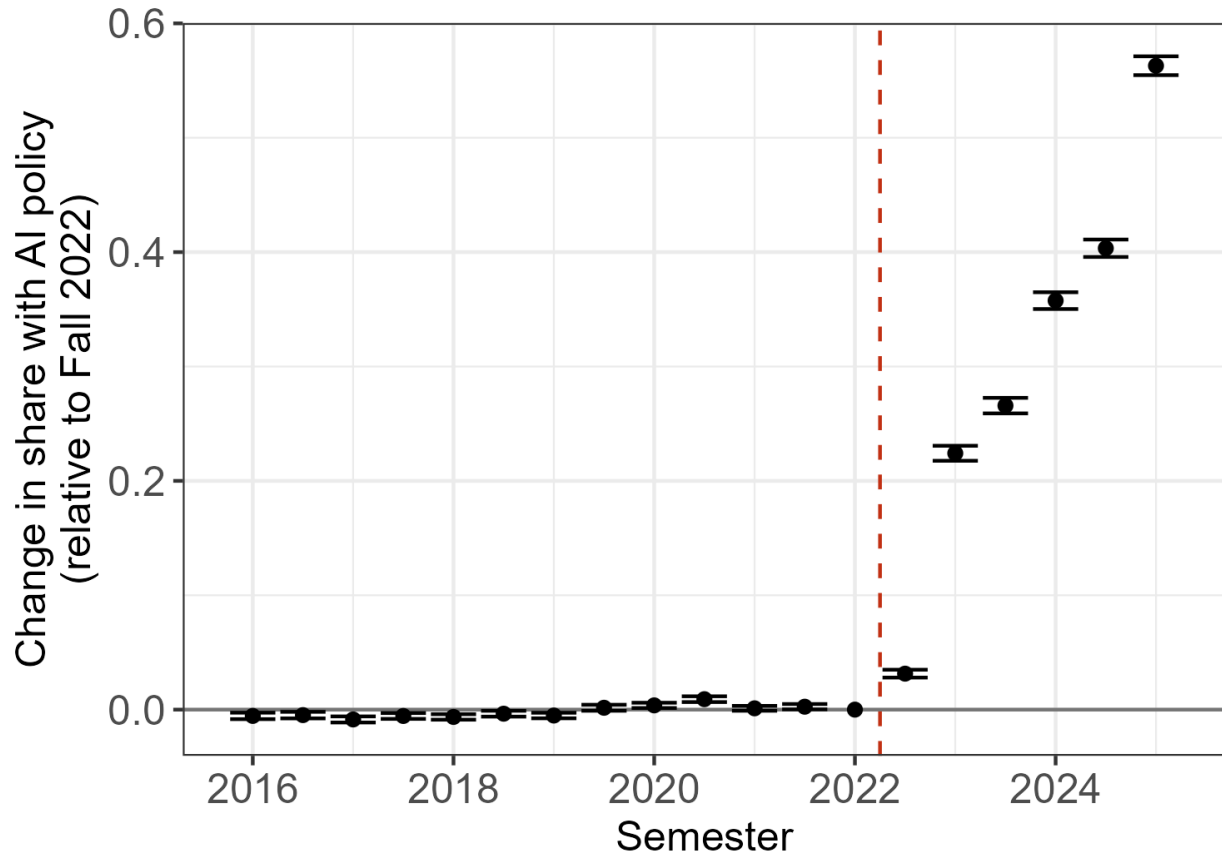
Notes: Grading weights from a sample of syllabi from 20 US colleges and universities. Observations at the course level; the dependent variable is the share of the course grade from homework, essays, or projects. Estimates from separate interrupted time series regressions for each field, controlling for course fixed effects. Fall 2022 is the omitted term. Time variable in academic years (thus, the 2022 point refers to the Fall 2022 semester, the 2022.5 point refers to Spring 2023, etc.). The dashed line indicates the public release of ChatGPT. Standard errors clustered at the course level.

Figure 7. Heterogeneity in grading weight changes by institution selectivity



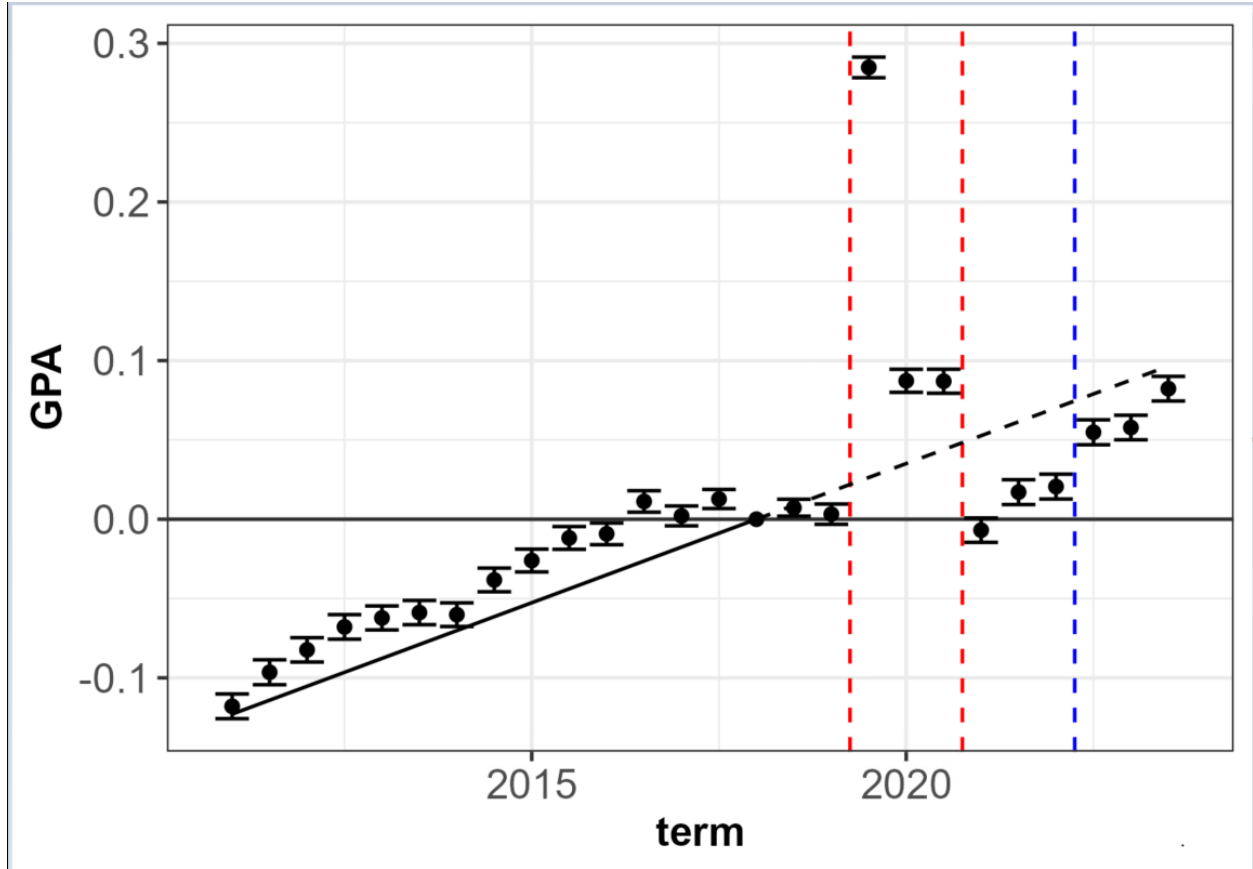
Notes: Grading weights from a sample of syllabi from 20 US colleges and universities. Observations at the course level; the dependent variable is the share of the course grade from homework, essays, or projects. Estimates from separate interrupted time series regressions for each institution category, controlling for course fixed effects. Fall 2022 is the omitted term. Time variable in academic years (thus, the 2022 point refers to the Fall 2022 semester, the 2022.5 point refers to Spring 2023, etc.). Selective institutions have admit rate below 60%. The dashed line indicates the public release of ChatGPT. Standard errors clustered at the course level.

Figure 8. Trend in AI policies



Notes: Estimates from interrupted time series with binary dependent variable (1 if the syllabus has an explicit AI policy), based on course-level observations from a sample of syllabi from 20 US colleges and universities. Fall 2022 is the omitted term. Time variable in academic years (thus, the 2022 point refers to the Fall 2022 semester, the 2022.5 point refers to Spring 2023, etc.). The dashed line indicates the public release of ChatGPT. AI policies pre-Fall 2022 are typically policies regarding the use of online translation tools in foreign language courses or related to AI-assisted exam administration during the pandemic. Standard errors clustered at the course level.

Figure 9. Grading trend around the release of ChatGPT



Notes: Estimates from regression of course grades (GPA points) on time dummies, controlling for course fixed effects, based on transcript data covering all four-year public universities in Texas. Observations at the student-course level, standard errors clustered at the course-term level. The red dashed lines mark the pandemic-exposed semesters (Spring 2020 and academic year 2020-21. Note that instruction during the 2020-21 academic year was primarily in-person at Texas institutions). The blue dashed line indicates the release of ChatGPT in Fall 2022.

Table 1. Common verb-object pairs for select fields

Field (1)	Verb (2)	Object (3)	% of tasks (4)	AI (5)	Robot (6)	Software (7)
Arts	develop	technique, concept, approach, appreciation	0.023	0.000	0.000	0.000
Arts	create	work, project, artwork, image	0.023	0.004	0.002	0.006
Arts	teach	music, lesson, skill, technique	0.018	0.000	0.000	0.001
Chemistry	perform	experiment, calculation, analysis, research	0.067	0.015	0.004	0.012
Chemistry	develop	technique, concept, perspective, principle	0.038	0.000	0.000	0.000
Chemistry	qualitative	analysis	0.035	0.000	0.000	0.000
Computer Science	develop	application, solution, system, program	0.019	0.003	0.000	0.020
Computer Science	process	spreadsheet, database, communication, graphic	0.019	0.000	0.000	0.003
Computer Science	create	application, page, visualization, plan	0.010	0.007	0.008	0.014
Economics	analyze	datum, problem, behavior, issue	0.045	0.124	0.028	0.050
Economics	develop	model, tool, theory, technique	0.033	0.019	0.004	0.003
Economics	apply	tool, method, analysis, model	0.032	0.038	0.009	0.006
Education	teach	lesson, skill, art, child	0.025	0.000	0.000	0.001
Education	learn	strategy, environment, disability, style	0.022	0.013	0.000	0.000
Education	develop	proposal, relationship, approach, philosophy	0.012	0.000	0.000	0.000
English	write	composition, requirement, analysis, fiction	0.063	0.000	0.000	0.000
English	read	text, literature, page, skill	0.034	0.000	0.000	0.001
English	develop	process, expertise, paper, question	0.023	0.000	0.000	0.001
Ethnic/Cultural Studies	examine	relationship, aspect, method, content	0.023	0.000	0.000	0.000
Ethnic/Cultural Studies	analyze	impact, representation, issue, study	0.021	0.003	0.000	0.002
Ethnic/Cultural Studies	develop	tool, appreciation, proposal, sense	0.016	0.004	0.000	0.001
Math	solve	equation, system, variety, model	0.194	0.000	0.002	0.005
Math	analyze	datum, graph, problem, relationship	0.028	0.091	0.021	0.038
Math	graph	system, analysis	0.025	0.007	0.000	0.000
Stats/Data Science	analyze	datum, output, set, result	0.078	0.214	0.052	0.088
Stats/Data Science	present	datum, result, concept, model	0.040	0.015	0.000	0.025
Stats/Data Science	test	analysis, power, model, application	0.017	0.001	0.000	0.002

Notes: The table lists the three most common verbs in each field of study and, for each verb, the (up to) four most common associated objects. Column 4 reports the share of all verb-object pairs represented by these selected pairs. Columns 5-7 multiply these shares by the technology weights for each technology class.

Additional figures and tables

Table 2. Excluded verb-object pairs highly distinctive of course descriptions

Verb	Object	Verb	Object
introduce	student	satisfy	requirement
provide	student	take	course
develop	skill	solve	problem
provide	opportunity	conduct	research
receive	credit	include	study
prepare	student	provide	experience
fulfill	requirement	include	analysis
provide	introduction	enable	student
provide	overview	offer	semester
develop	understanding	meet	requirement

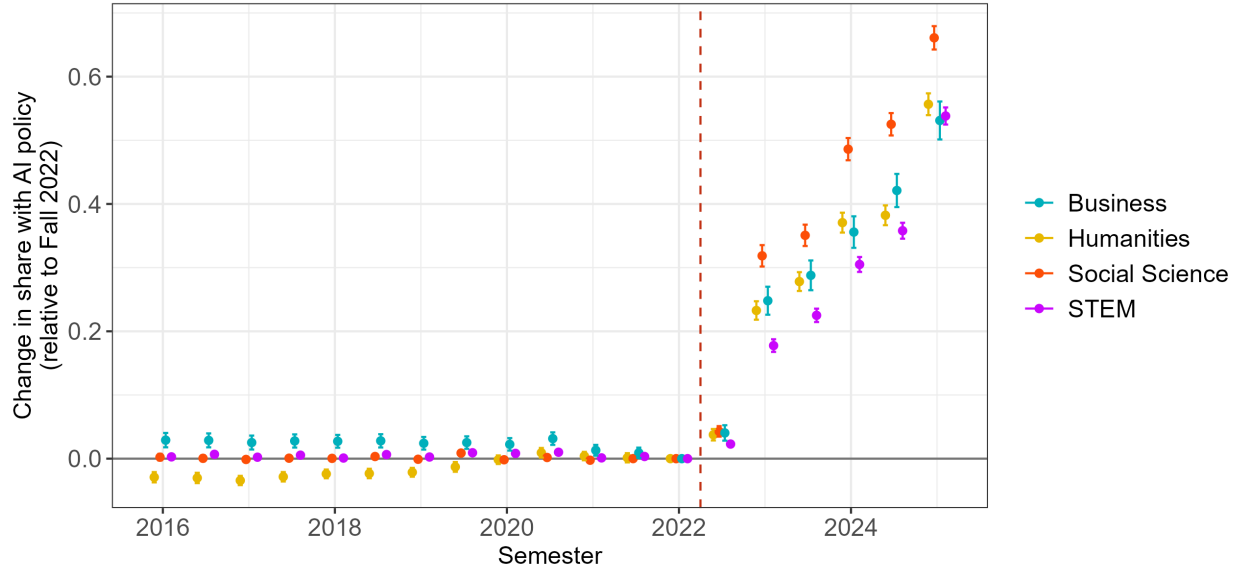
Notes: The table shows the 20 verb-object pairs (of the 100 excluded from the pair dictionary) that are most distinctive of course catalogs. The pairs are ranked by the chi-square statistic comparing their frequency shares in course descriptions to their frequency shares in patents.

Table 3. Summary of task counts in course description data

	Catalog [1]		Occupation [2]		Patent [3]		Occupation + patent [4]	
	Unique	Total	Unique	Total	Unique	Total	Unique	Total
1 - full corpus	248,088	3,759,797	12,092	25,697	122,027	461,387	129,691	487,084
2 - remove uncommon pairs	112,735	3,576,337	12,092	25,697	30,180	348,955	31,819	367,463
3 - remove catalog-specific pairs	112,645	3,078,562	12,058	25,123	30,161	348,493	31,788	366,428
4 - project onto patent/occupation pairs	10,017	719,479	12,058	25,123	30,161	348,493	31,788	366,428
5 - remove stopwords	8,110	452,331	10,942	21,564	26,150	254,375	27,612	269,258

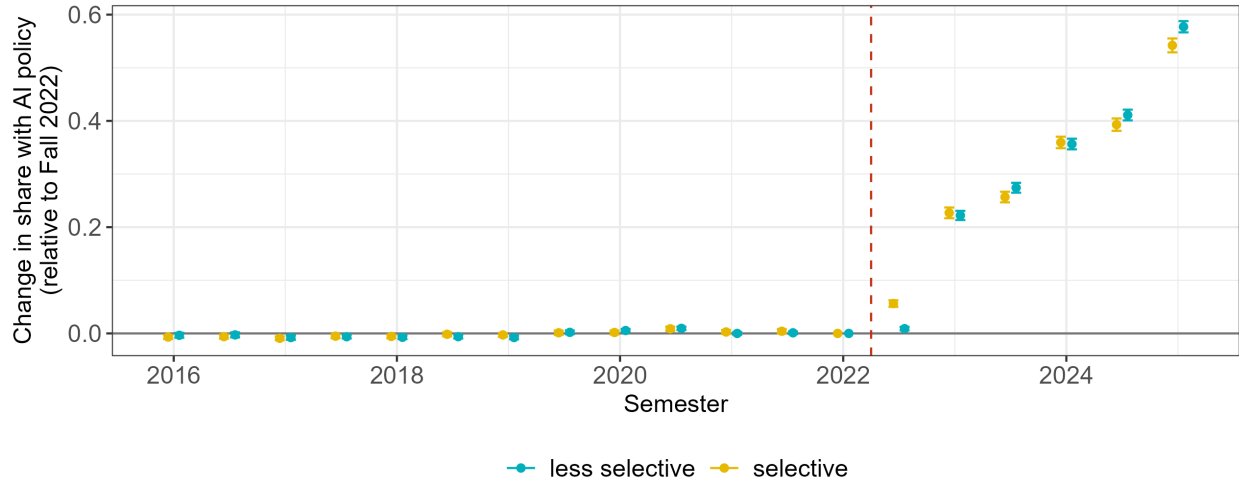
This table reports counts of unique and total verb-object pairs at each stage of data processing. Counts are shown for course descriptions (1), occupation tasks from O*NET (2), patents (3), and the union of the O*NET and patent corpora. Row (1) covers the full corpora. Row (2) excludes extremely uncommon pairs (fewer than three appearances) in the catalog and patent data (not applied to O*NET, which is sparser and has higher signal). Row (3) removes 100 pairs that are highly distinctive of course descriptions. Row (4) projects the catalog pairs onto the dictionary of pairs found in either O*NET or patents. Row (5) removes stopwords.

Figure 10. Heterogeneous AI policies by field of study



Notes: Estimates from separate interrupted time series for each field category with binary dependent variable (1 if the syllabus has an explicit AI policy), based on course-level observations from a sample of syllabi from 20 US colleges and universities. Fall 2022 is the omitted term. Time variable in academic years (thus, the 2022 point refers to the Fall 2022 semester, the 2022.5 point refers to Spring 2023, etc.). The dashed line indicates the public release of ChatGPT. AI policies pre-Fall 2022 are typically policies regarding the use of online translation tools in foreign language courses or related to AI-assisted exam administration during the pandemic. Standard errors clustered at the course level.

Figure 11. Heterogeneous AI policies by institution selectivity



Notes: Estimates from separate interrupted time series for each field category with binary dependent variable (1 if the syllabus has an explicit AI policy), based on course-level observations from a sample of syllabi from 20 US colleges and universities. Fall 2022 is the omitted term. Time variable in academic years (thus, the 2022 point refers to the Fall 2022 semester, the 2022.5 point refers to Spring 2023, etc.). The dashed line indicates the public release of ChatGPT. AI policies pre-Fall 2022 are typically policies regarding the use of online translation tools in foreign language courses or related to AI-assisted exam administration during the pandemic. Standard errors clustered at the course level.